

Auteur: Odia Kibor  
Studierichting: Informatica  
Oefeningenreeks 1B nr. 37.20

$$24i - 32$$

$$= (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 &= -32 \\ 2xy &= 24 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 &= -32 \\ xy &= 12 \\ (xy > 0) \end{cases} \Rightarrow \begin{cases} x^2 - y^2 &= -32 \\ x^2(-y^2) &= -144 \end{cases}$$

$$\lambda^2 + 32\lambda - 144 = 0$$

$$D = 1600$$

$$\lambda_{1,2} = \frac{-32 \pm 40}{2}$$

$$\lambda_1 = 4 \text{ en } \lambda_2 = -36$$

$$x^2 = 4 \Rightarrow x = \sqrt{4}$$

$$-y^2 = -36 \Rightarrow y = \sqrt{-36}$$

$$\text{Oplossing: } 2 + 6i \text{ of } -2 - 6i$$

## References